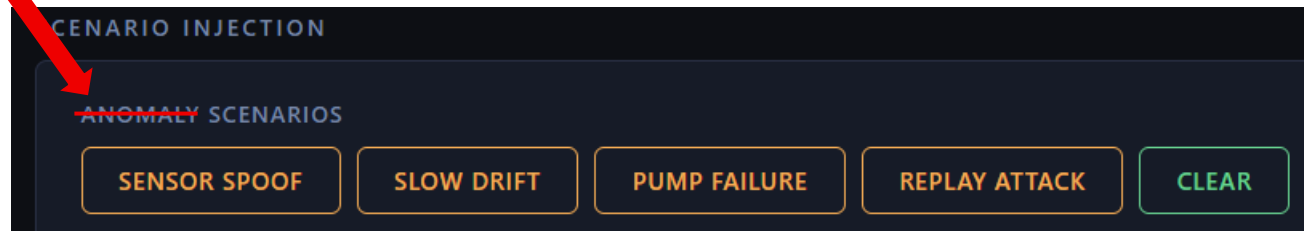


Remove the word "Anomaly". Should just be "Scenarios": PILOT is NOT anomaly detection



SCENARIO INJECTION

ANOMALY SCENARIOS

SENSOR SPOOF

SLOW DRIFT

PUMP FAILURE

REPLAY ATTACK

CLEAR

OPERATING MODE

NORMAL

RECIRCULATION

SLOW DRIFT ATTACK

ATTACK

WATCH FOR

DETECTION

AT-101 (chlorine residual analyser) reading is being incremented **+0.02 mg/L per tick**. Real concentration remains at setpoint (~2.0 mg/L). The manipulation is designed to be invisible to any single-point check — it looks like normal process variation.

AT-101 will slowly climb above 2.0 mg/L over the next ~10–20 ticks. **CT-101 conductivity** (which tracks real concentration with a lag) will stay flat — revealing that the reported concentration has no physical basis.

LAYER 2 — RESIDUAL SCORER

Sustained AT-101 residual growing each tick; secondary envelope (CT-101) contradicts

Add simple clarifier:

“Reported chlorine increase is not corroborated by expected downstream conductivity response, dosing behaviour, hydraulic flow conditions, or expected temporal process evolution.”

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All primary and secondary readings are frozen to a snapshot taken at the moment the attack began. Tick number and timestamp continue advancing, but every sensor value is identical each tick. An operator watching individual readings would see nothing unusual.

WATCH FOR

P-102 dosing is still active — AT-101 concentration should be rising as chemical is injected, but it will not move. CT-101 conductivity should track concentration changes but will also stay flat. The frozen values are the tell.

DETECTION

LAYER 2 — RESIDUAL SCORER

Concentration flat despite active dosing; residual grows against expected rise

LAYER 3 — COUNTERFACTUAL

Replay hypothesis: if dosing were truly active, AT-101 and CT-101 would be rising

Add simple clarifier:

“Although timestamps and control execution continue advancing, downstream process variables fail to evolve within expected causal and temporal bounds for an actively dosing system.”